Software Requirements Specification

for

Dress Code Violation (DCV) Management System

Version 1.4 approved

Prepared by Syntaxxed

Asia Pacific College

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Revision History

Name	Date	Reason For Changes	Version
Maria Theresa Carreon	December 09, 2024	System features were not separately written	1.1
Daniel Bucayan	December 12, 2024	Added more features in the system features	1.2
Nelson Mabini	December 12, 2024	Added Appendices A, and slight numbering errors	1.2
Michael Patrick Escalambre	December 13, 2024	Updated External Interface Requirements	1.2
Daniel Bucayan	February	Updated prototype images and some grammar in	1.3

	17, 2025	the document	
Kim Sung Tae	February 17, 2025	Updated Web prototype images added new images for AI model and rearranged the numbering to 3.1.1 - 3.1.3	1.4

1. Introduction

1.1 Purpose

This document outlines the software requirements for our Dress Code Violation (DCV) management system. It is designed to automate issuance and management of dress code violations implemented at Asia Pacific College (APC). The system utilizes technology to simplify processes, by using smartphones and computers, it effectively removes manual errors, reduces inefficiencies, and ensuring consistent enforcement of dress code policies.

1.2 Document Conventions

This document follows a set of typographical and formatting conventions to ensure clarity and consistency:

- **Bold Text**: Used to emphasize important terms, section titles, and key concepts within the document to draw attention to critical information
- **Numbered Lists**: Used to present a sequence of steps or prioritized information.
- **Bulleted Lists**: Provides categorized items for quick reference.
- **Table Format**: A way to show data in an organized manner or comparisons for easy readability.

1.3 Intended Audience and Reading Suggestions

This document is intended for users, developers, the client, and the documentation writers to provide context and information regarding the functions and processes with regards to the product.

Users – Individuals who will interact with the system, each with their respective roles in the functionalities of the system.

1. Head of Discipline Office - Oversees the management of dress code violations, reviews reports, and ensures consistent enforcement of policies within the organization.
2. Security Guard - Identifies dress code violations, issues violation slips using the system and ensures adherence to dress code policies.
3. Student - Receives notifications of violations, acknowledges issued violation slips, and submits appeals if necessary, promoting accountability and compliance.

Developer: The technical team responsible for designing, building, testing, and maintaining the DCV Management System to ensure it meets functional and performance requirements.

Project Client: The client commissioning the system, providing requirements, feedback, and approval during the development process, which is also the Head of Discipline Office.

Documentation Writer: The group responsible for creating clear and comprehensive documentation, including user manuals and technical guides, to support all stakeholders in effectively using and understanding the system.

1.4 Product Scope

The **Dress Code Violation (DCV) Management System** is a solution specifically designed to help the disciplinary office in their record management, its primary purpose is to ensure the enforcement of policies, reduce errors and to enhance operational efficiency. This system will also strengthen communication between students, staff, and administrators, fostering accountability and transparency throughout the process.

Student Portal for Real-Time Updates: Allows students to view their violations, submit appeals, and track their status through a secure online interface.

Head D.O Portal for Violation Management. Allows the Head D.O to manage and review received violation records.

Violation Analytics and Reporting: Offers detailed insights into dress code violations, enabling data-driven decisions to improve policy enforcement.

Administrator Dashboard for Comprehensive Monitoring: Provides administrators with tools to manage records, generate reports, and assess compliance trends.

The project aims to align with the institution's goals of maintaining a disciplined and professional learning environment. It also integrates with Asia Pacific College's (APC) broader strategy to embrace digital transformation and enhance the overall experience for its stakeholders.

2. Overall Description

2.1 Product Perspective

The project was initiated by the developers to **replace the existing manual blue slip system** of Asia Pacific College. This is done to address several flaws of the existing manual system for blue slip issuance, such as the inefficiency of the process, the formation of queues at the entrance, integrity issues, and the unsustainability of the current system. The proposed project will utilize QR code scanning for efficiency of credential retrieval of students, along with an AI that detects dress code violations, and several more functions to aid in managing violation records efficiently.

2.2 Product Functions

The project consists of a mobile application, web portal, and an AI service which contains the following functions:

Mobile Application:

- **QR Code Scanning** – Scan students' QR Codes to get the credentials needed in issuing a violation slip.
- **Generate Violation Slips** – Read, create, and submit violation slip after the retrieval of necessary information.

Web Portal:

- **File Appeal** – Enter information and reason for filing an appeal on a given violation.
- **Violation Management** – Confirm and record submitted violation slips without an appeal.
- **Appeal Management** – Confirm and update violation slips with approved or denied appeal requests.
- **Analytical Report Generation** – Request and generate an analytical report based on a specified timeframe or violation.

AI Service:

- **Violation Detection and Identification** – Detect students' attire and determine if there are any violations on the dress code policy of APC.

2.3 User Classes and Characteristics

The project is anticipated to be used by three users: **Students**, **Security Guards**, and the **Head of Discipline Office**. Table 1 provides detailed information regarding the user classes of our product, including how they utilize the system and their pain points.

TABLE I
USER CLASSES AND CHARACTERISTICS

User	Characteristics
Student	These types of users are students who were issued violation slips. They take part in the process by being able to view issued violations and filing an appeal, if they want to contest the issued violation. Pain points: • Queues formed in the entrance can obstruct their way to class. • It takes time to fill out violation slips.
Security Guard	These types of users are Security Guards who are the ones responsible in issuing violation slips to students who do not follow the dress code policy. They scan students' QR codes and generate violation slips through the mobile application. Pain points: • Issuing violation to students take time. • Issued violation slips are personally brought to the Discipline Office.
Head of Discipline Office	This type of user is the Head of the Discipline Office who is responsible for managing and recording the violation slips issued by the security guards. They review issued violations through the web portal and review appeal requests submitted by students. They can also request analytical reports to gain insights and possibly make updates with the dress code policy of APC. Pain points: • Blue slips are recorded manually through MS Excel, which takes several hours to do. • Blue slips are prone to errors, and the handwriting is illegible sometimes. • Blue slips are unsustainable. • Blue slips are not logged by the security guard and the Discipline Office.

2.4 Operating Environment

The project will be operated in the following environment:

Platform - A web-based browser for the portal, which can be accessed by Chrome, MS Edge, etc., and an Android OS for the mobile application utilized in generating violation slips.

Operating System – The server environment will be hosted in the Hostinger platform, which utilizes a Linux-based operating system.

Database – The system will use PhpMyAdmin for the database to store data such as student data, violation slip data, and analytical report data.

Additional Components

- **Hosting Platform** – The web application of the system will be hosted on Hostinger.

2.5 Design and Implementation Constraints

The design and implementation of the project is subject to the following constraints, which limit the developers' options:

- **Development Timeframe:** The project is expected to be finished within a limited timeframe, which limits the developers to focus on the core aspects of the system.

- **School Dress Code Policy:** The project is only implemented in Asia Pacific College (APC), which only limits the scope of the system to APC's dress code policies.

- **Project Integration:** The project will be integrated with the existing information system of APC, which will limit developers to making the project compatible with it.

- **Security Considerations:** The project will have security implementation within it to prevent or mitigate risks such as DoS attacks, SQL injections, etc. It will also adhere to the Data Privacy Act of 2012.

2.6 User Documentation

Several documentations will be given along with the system to guide users on the functionality of the product and give context regarding the system. The documentation is the following:

User Manual

The user manual provides step-by-step instructions in using the services the systems offer and a basic overview of the system, such as the following:

- **Introduction** - DCVMS (Dress Code Violation Management System) is an administrative panel designed for reviewing student dress code violations. It allows administrators to log in, review reported violations, and decide whether to approve or reject them. This manual provides step-by-step instructions to navigate and use the system effectively.
- **Getting Started**
    - System Requirements
        - Stable Internet connection
        - Modern web browser (Chrome, Firefox, Edge)
        - Valid administrator login credentials (this is only available if the ITRO provides it for you)
    - Accessing DCVMS
        - Open the web browser
        - Enter the username and password provided
        - Click the Login
        - If the credentials are correct, it will redirect you to the dashboard
- **Dashboard Overview**
    - **A**fter logging in the dashboard will provide the following
        - Graphs of the violations
        - To be reviewed cases
        - Search Bar
        - Navigation Menu
- **Reviewing Violations**
    - Click the Review button
    - Choose whether to approve or reject the violation
    - Click Submit Button

- **Troubleshooting**
  - Unable to log in
    - Make sure the username and password you entered are provided by the school.
    - If the problem keeps occurring, please contact the ITRO.

  - Violation details are not loading
    - Refresh the page and check your internet connection. Contact the ITRO if the issue persists.
  - Cannot approve or reject violations
    - Make sure the account you entered has permission to approve or deny appeals. If the problem continues, please contact the ITRO.

FAQs

The Frequently Asked Questions (FAQs) portion will address common questions or queries regarding the system, such as:

- How do I file an appeal?
- What is the deadline for filing an appeal?
- How can I view the details of my violation slip?
- What happens if I get three or more violation slips?
- Can I access the portal through my phone?
- Can I edit my submitted appeal in case of errors?
- What do I do if the AI doesn’t detect the violation?

2.7 Assumptions and Dependencies

For proper and successful implementation of the system, the developers used the following assumptions and dependencies as basis:

Assumptions:

- **Detailed Dress Code Policy:** It is assumed that APC already has a detailed dress code policy, and that everyone is notified in case of updates and changes made to the said policy.

- **User Devices:** It is assumed that each user class of our system already has their own devices to properly access the system.

- **Framework Acceptance:** It is assumed that the framework of our system, Django, will be approved and accepted by the APC.

- **Project Scope:** The project mainly focuses on the dress code policy and its violations only.

- **Hosting and Deployment:** It is assumed that the system will be hosted under APC's cloud environment and will be deployed in the cloud once completed.

Dependencies:

- **System Integration:** The system might need to be integrated with APC's current information system to access data seamlessly that is needed by the system.

- **Data Availability:** Student data of APC is needed for some of the systems to function. Dataset for the AI's training would also be needed, if there are any available. Otherwise, the dataset would be created.

3. External Interface Requirements

3.1 User Interfaces

The Web-based application for Dress Code Violations (DCV) Management system gives the user access to a user-friendly interface with easy access to buttons, allowing for **seamless navigation through the system. It provides the user with a dashboard to monitor and effectively manage administrative tasks. Only the Head of the Discipline Office is authorized to manage these tasks and serve as the sole administrator for the web application.**

3.1.1 AI Model General Design

For the AI model, there will be cameras set up in front of each entrance of the school which will detect students who violate the dress code of Asia Pacific College.

Sample Detection for ID

Sample Detection for Inappropriate Attire (Shorts)

3.1.2 Mobile Application General Design

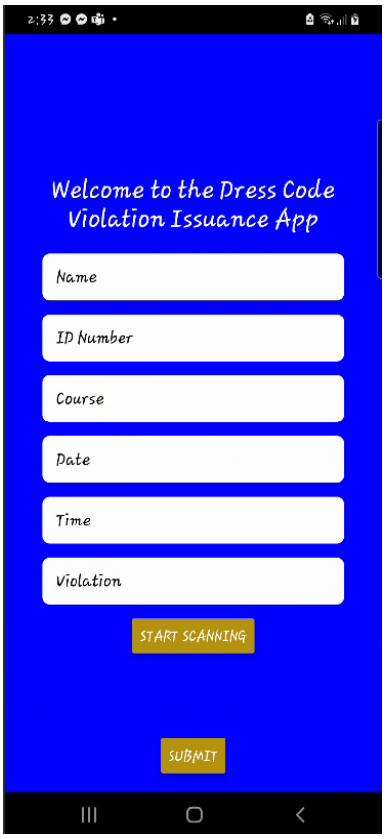
For the mobile application, only the security guards are authorized to access and manage this part of the system as they are responsible for issuing and scanning the QR codes of the students. The application's interface has been designed to be user-friendly, ensuring easy accessibility and navigation specifically tailored to the needs of the security guards.

Mobile Application Interface

- User-friendly interface with easy access to buttons that functions well and has a responsive design that ensures the user can navigate the application at ease without any confusion.

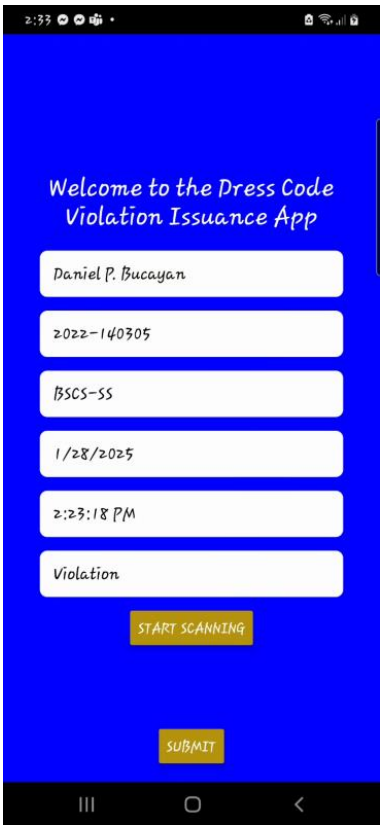
Scanning Interface

Application Home Screen



Violation Drop-Down Menu

Automated Input



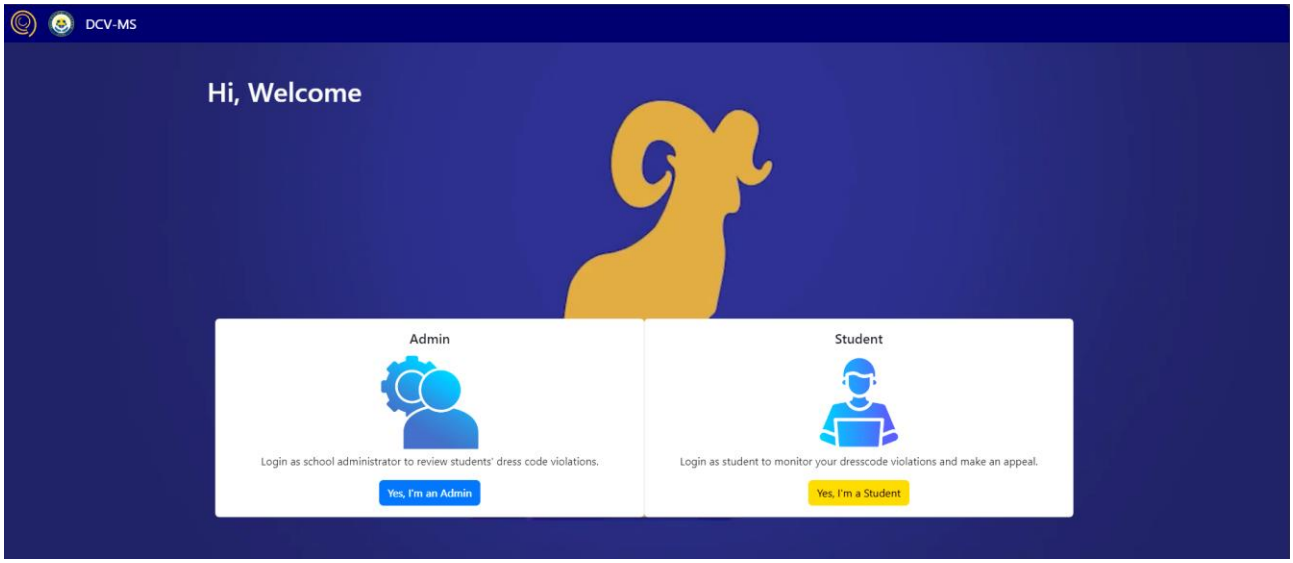
Submitting Blue Slip

3.1.3 Web Portal General Design

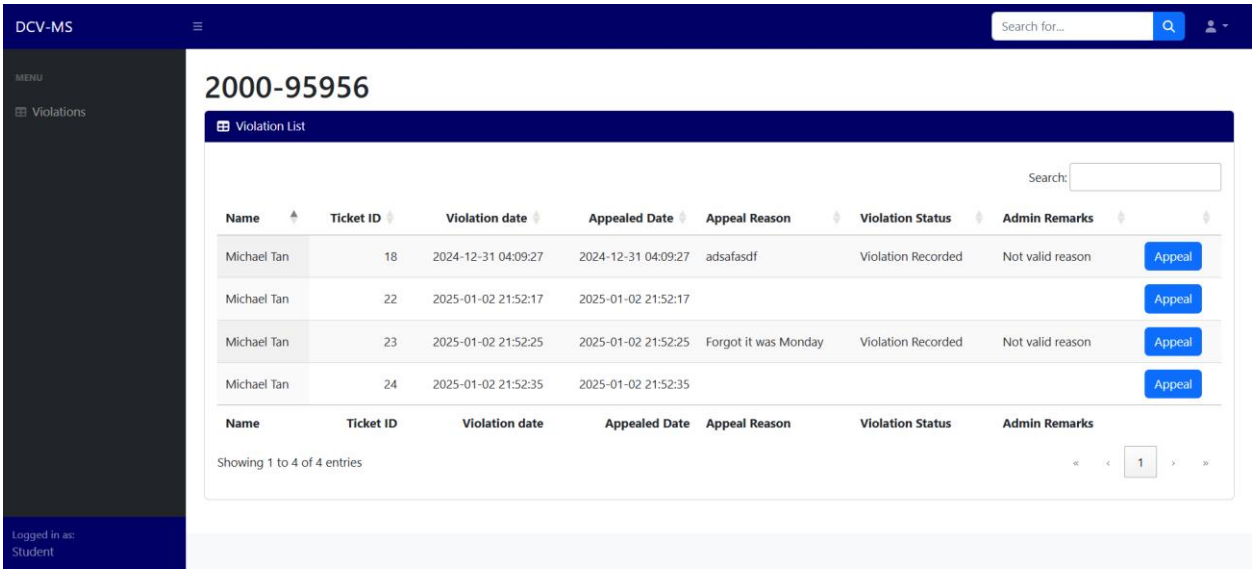
Web Portal/ Interface

- Can be accessed through browsers (**Chrome, Edge, Opera, Firefox, Brave).**
- Responsive **design for easy compatibility and navigating.**

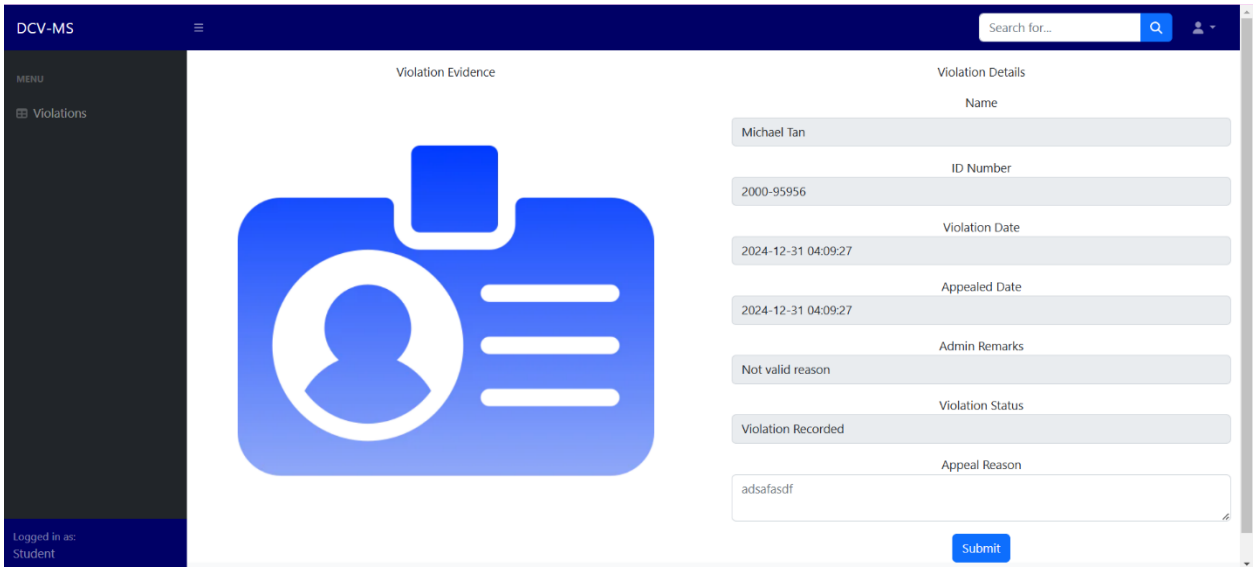
Login



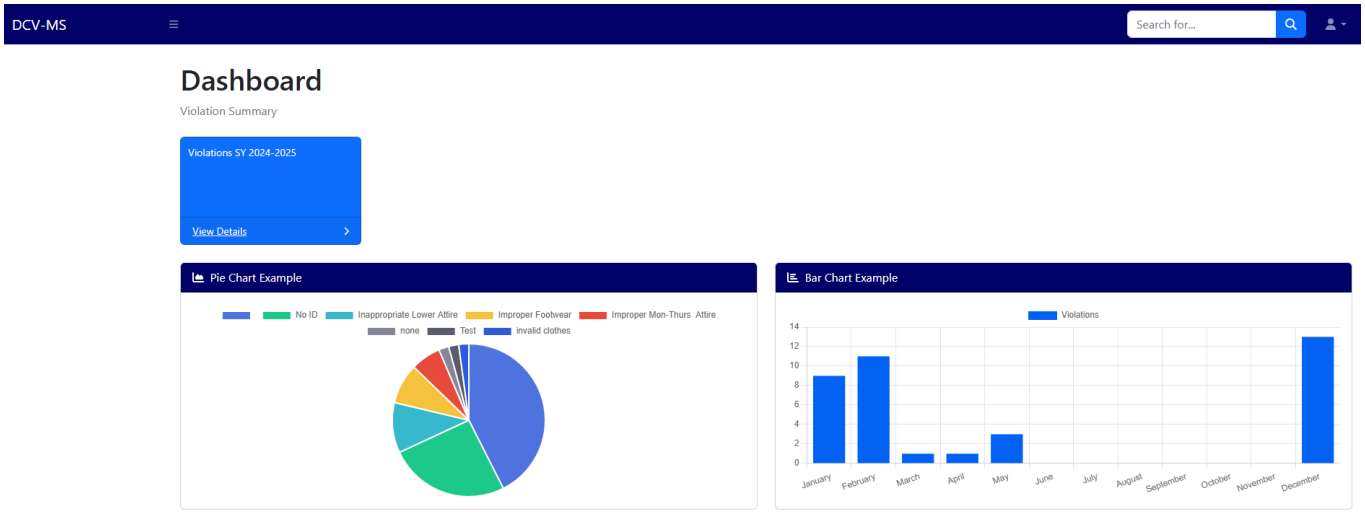
Student Homepage

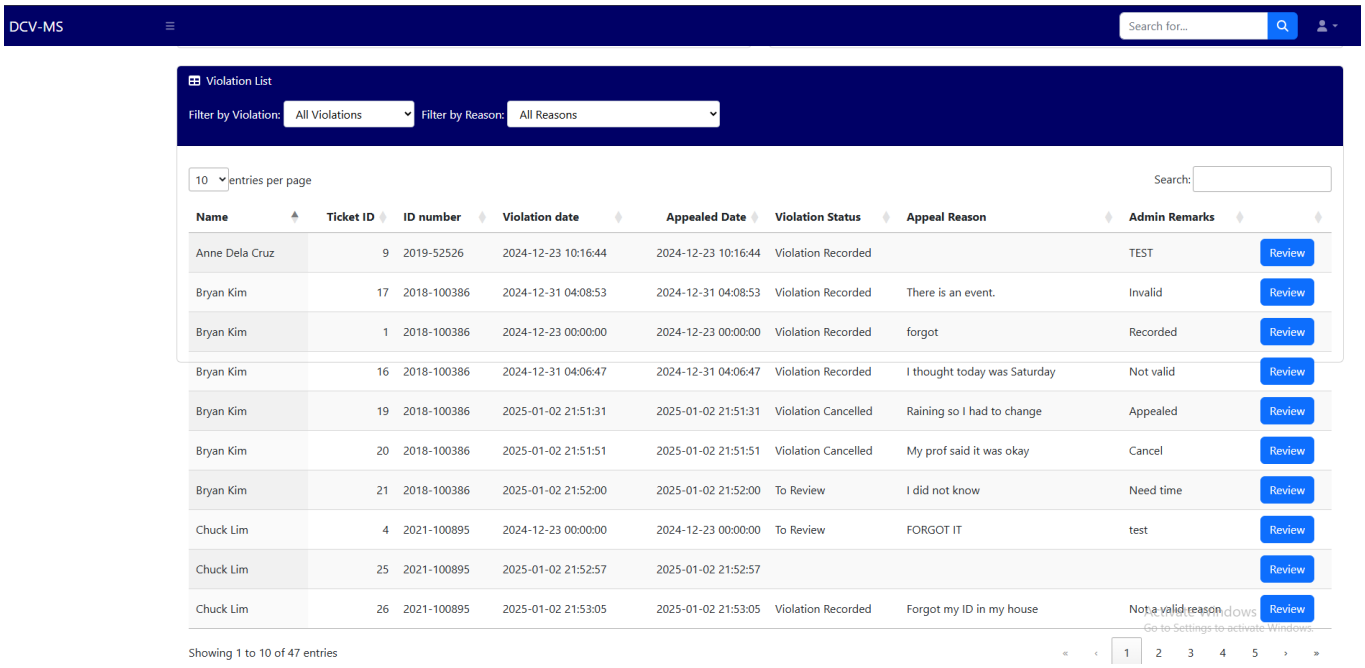


File an Appeal (Students)

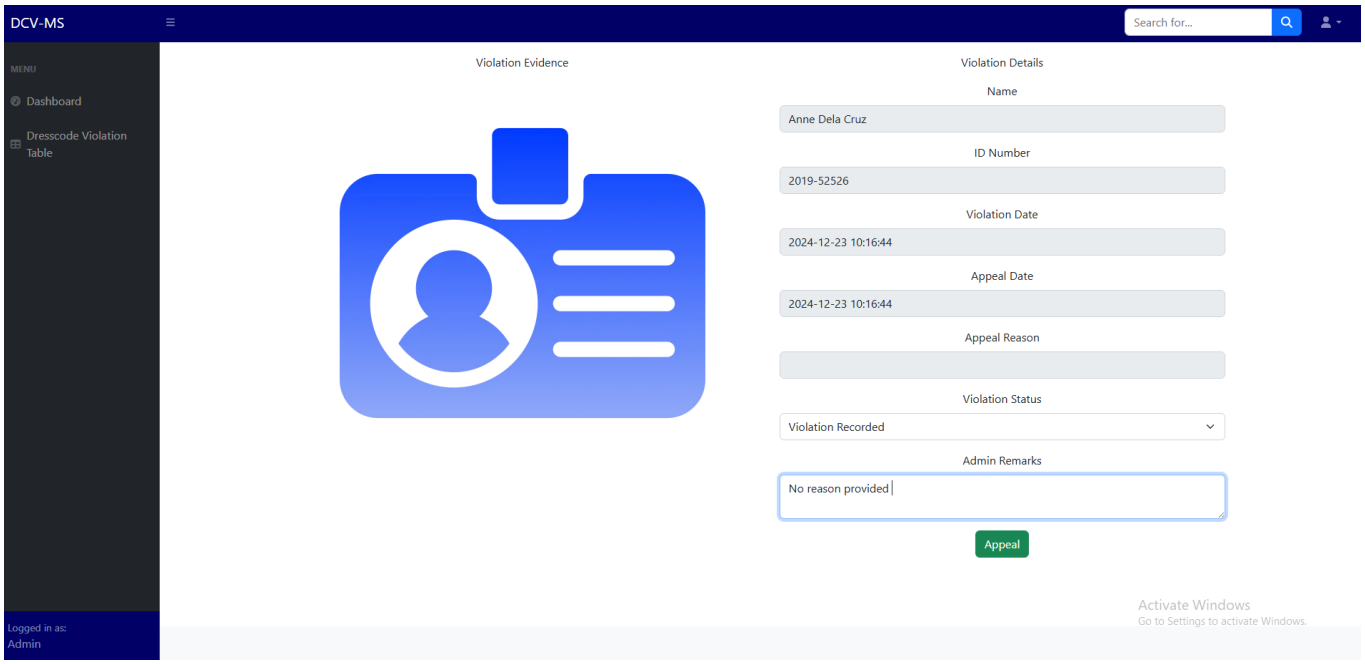


Admin Homepage

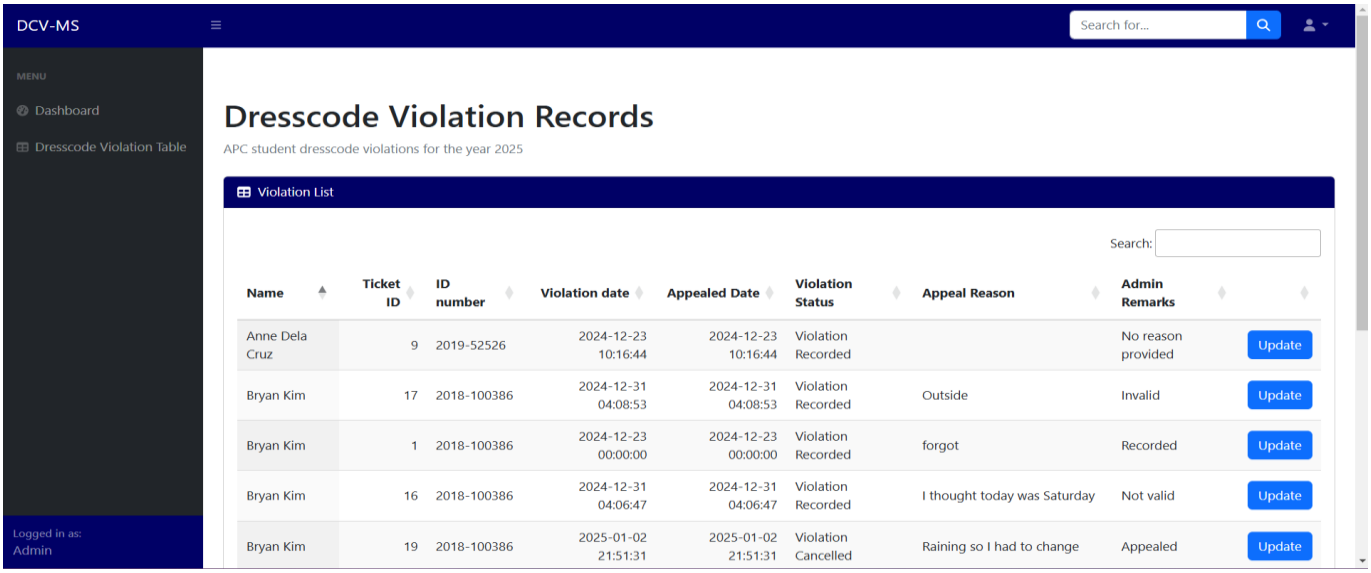




Reviewing an Appeal (Admin)



Violations on Record



3.2 Hardware Interfaces

The system interacts with the following hardware components needed such as:

Clients Device (Head of Discipline Office and Students):

- Desktop for the user to be able to access the web portal, and smartphones to be able to access the mobile application.

Minimum Requirements:

- **Desktop:**
  - Processor: Intel Xeon or Higher
  - Memory: 8-16 Gb of ram
  - Storage: 500Gb to 1Tb of HDD/ SSD Storage For handling and containing data.
  - Operating System: Windows and Linux
  - Server Software: MySQL and PHP
- **Backup Server**
  - Regularly backup data to ensure any issues can be recovered quickly.

Clients Device (Security Guard):

- **Smartphone:**
  - Operating System: Android 6 or higher.
  - Memory: 6 Gb of ram
  - Storage: 1 Gb of free space
- **Server Infrastructure**
  - Hosted on Hostinger Platform
    - Memory: 16 Gb of ram
    - Disk Space: 400 Gb HDD or SSD
    - Processor: Intel Xeon or Higher
    - Network: 5 Gbps ethernet connection or higher

3.3 Communications Interfaces

The communication interface ensures that the software will interact with the external systems and services:

Internet Connectivity:

- **Minimum Bandwidth:** A minimum internet connection of 2 Mbps is required to ensure optimal performance.

Push Notification Services:

- Support sending real-time notifications to users for updates, alerts, and reminders.

Real-Time Updates

- Enables instant feedback and notifications through communication channels.

4. System Features

The Dress Code Violation (DCV) Management System includes features that enhance efficiency and streamline the process of the Disciplinary Office's dress code violation issuance. High Priority features focus on core functionalities such as Role based access control, a mobile app for blue slip issuance, a web portal for the disciplinary office and students, and cloud hosting. Medium priority features include automated notifications, trends and report analysis and data backup for recovery. Feedback collection for future updates is included in the low priority feature. These priorities ensure the system delivers immediate value while remaining adaptable for future improvements.

4.1 Violation Identification: DCV-REQ1

- **Description:** The AI Camera determines the violation of the student.
- **Priority:** High Priority
- **Pre-Condition:**
  - Student commits a dress code violation.
- **Stimulus:** AI Camera determine the student's dress code violation.
- **Response:** The violation will be transferred to the mobile app that will be accessed by the Security Guard.
- **Functional Requirement:**
  - The AI camera shall analyze live video feeds to detect dress code violations based on the latest student handbook.

- The AI model shall achieve an accuracy rate of at least TBD% for detecting violations.
- The system shall ensure that the transmission occurs within TBD seconds of violation detection.

4.2 QR Code Scanning: DCV-REQ2

- **Description:** Mobile App scans the QR code of the student who violates the Dress Code Policy.
- **Priority:** High Priority
- **Pre-Condition:**
  - Student has his/her ID and QR Code for scanning.
- **Stimulus:** The security guard scans the QR code of the student using the mobile app.
- **Response:** The mobile app will gather the information imbedded to the QR code of the student.
- **Functional Requirement:**
  - The security guard shall scan the Student's QR code according to the detected violator of the AI camera.
  - The app shall validate QR code data on the student database and display the student's details.

4.3 Violation Issuance Management: DCV-REQ3

- **Description:** Security guards manage issuance of dress code violation for the student who violates dress code policy.
- **Priority:** High Priority
- **Pre-Conditions:**
  - A Student commits a dress code violation.
  - The QR code is scanned to gather the information.
- **Stimulus:** The system processes the details received from the QR Code and the violation details.
- **Response:** The Security Guard issues the violation slip to the violator.
- **Functional Requirement:**

  - The mobile app shall allow security guards to issue dress code violations after scanning a student's QR code.

4.4 Read Violation Record: DCV-REQ5

- **Description:** The student record is checked whether the student has already been issued a violation slip before.
- **Priority:** High Priority
- **Pre-Conditions:**
  - A Student commits a dress code violation.
  - The QR code is scanned to gather the information.
- **Stimulus:** The system processes the details received from the QR Code and the violation details.
- **Response:** The system will read the record of the student's violation to the database and identify any existing violation slip issued.
- **Functional Requirement:**

  - The mobile app shall allow security guards to issue dress code violations after scanning a student's QR code.

4.5 Create Violation: DCV-REQ4

- **Description:** Security guards create a violation slip for the student who violated the dress code policy.
- **Priority:** High Priority
- **Pre-Conditions:**
  - A Student commits a dress code violation.
  - The QR code is scanned to gather the information.
  - The record of possible past issued violations is read.
- **Stimulus:** The security guard issues dress code violation to the violator.
- **Response:** The system will record the violation to the database and notify the student about the violation.
- **Functional Requirement:**

  - The mobile app shall allow security guards to issue dress code violations after scanning a student's QR code.

4.6 Appeal Request: DCV-REQ6

- **Description:** The student files an appeal and is submitted to the Discipline Office for review.
- **Priority:** High Priority
- **Pre-Conditions:**
  - Students must have a pre-existing violation. The student has access to the web portal to submit an appeal.
  - The violation is eligible for appeal as per school policy.
- **Stimulus:** The student files an appeal for the violation.
- **Response:** the system will save the appeal so that the Discipline's office Head can review the appeal and decide whether to approve or deny it.
- **Functional Requirement:**
  - Role-based access control will be used by the system to guarantee that only authorized users have access to specific features.
  - Students shall use the portal to view violations, file appeals, and check appeal statuses.

4.7 Violation Management: DCV-REQ7

- **Description:** Allows the Head of the Discipline's Office to manage issued violations that don't have appeal requests.
- **Priority:** High Priority
- **Pre-Conditions:**
  - Violation is submitted to the system.
  - The Head of the Discipline Office receives the violation slips.
- **Stimulus:** The Disciplinary Office Head reviews the violation slips issued from the student.
- **Response:** The Disciplinary Office Head records the violation slips for analytics purposes.
- **Functional Requirement:**
  - Role-based access control will be used by the system to guarantee that only authorized users have access to specific features.
  - The web portal shall allow the Head of the Discipline's Office to view and manage violations.

4.8 Appeal Management: DCV-REQ8

- **Description:** Allows the Head of the Discipline's Office to manage issued violations that have appeal requests.
- **Priority:** High Priority
- **Pre-Conditions:**
  - Violation is submitted to the system.
  - The Head of the Discipline Office receives the violation slips.
- **Stimulus:** The Disciplinary Office Head reviews the appeal request of the student.
- **Response:** The Disciplinary Office Head decides whether to approve or deny the appeal of the student based on the given reason for appeal.
- **Functional Requirement:**
  - Role-based access control will be used by the system to guarantee that only authorized users have access to specific features.
  - The web portal shall allow the Head of the Discipline's Office to view and manage violations, review appeals, and decide whether to approve or deny the appeal/s.

4.9 Update Violation Slip Record: DCV-REQ9

- **Description:** Allows the Head of the Discipline's Office to update recorded violations based on the decision regarding the validity of the appeal.
- **Priority:** High Priority
- **Pre-Conditions:**
  - The Head of the Discipline Office reviews the appeal request.
- **Stimulus:** The Disciplinary Office Head approves or denies the appeal request of the student.
- **Response:** The violation slip of the student is updated, and the students will be notified of the decision.
- **Functional Requirement:**
  - The web portal should update the records in the database and should notify the student.

4.10 Analytical Report Generation: DCV-REQ10

- **Description:** The Head of Disciplinary Office can generate an analytical report to be able to see the pattern of the violations for a specific period.
- **Priority:** Medium Priority
- **Pre-Conditions:**
  - The discipline office head must be logged into the system with appropriate access rights.
  - The dress code violation data must be stored and available in the systems for analysis.
- **Stimulus:** The system will record violations and violators.
- **Response:** The discipline office Head can generate a report to see the pattern or trends regarding the dress code violation.
- **Functional Requirement:**
  - Role-based access control will be used by the system to guarantee that only authorized users have access to specific features.
  - The web portal shall allow the Head of the Discipline's Office to generate a report for a selected time frame.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

To ensure that our system runs smoothly and flawlessly, the following must be always met.

- Storage Capacity
  - The system must have enough storage to meet the project’s needs.
  - Must have enough storage space to store all the data.
  - Must be able to withstand the whole community in APC.
- Execution Time
  - The AI camera should be able to detect and execute in less than 10 seconds.
  - The QR code must be read in less than 5 seconds.
  - Generation of a violation slip should be retrieved in less than 5 seconds.
  - Analytic reports should be saved in less than 5 seconds.

5.2 Safety Requirements

The safety requirements are needed to ensure that the user and the system operate without fail.

- System Maintenance
  - Monthly maintenance will be done to ensure that the system will not go down
  - The users will be notified of the maintenance and their dates.
  - The downtime during the maintenance will be less than an hour.
- Data Recovery
  - In case of system failures, the system will be using backup data to prevent data being lost.
  - The recovery must be accomplished within an hour.

5.3 Security Requirements

The system must follow these requirements to ensure that the sensitive data will be always secured.

- Data Encryption
  - All the data collected by the AI will be encrypted to protect against unauthorized access
- Access Control
  - The system will be implementing strict access control to limit who can access the system and its data.
  - Will be using multi-factor authentication for added security measures.
- Regular Updates and Patching
  - The system will have a regular update and patch to ensure that there will be no vulnerabilities and threats to the system and its data.
- Audit and Monitoring
  - Implementation of monthly audits and continuous monitoring will be done after deployment to ensure that there aren't any suspicious activities or anomalies
- User Training
  - Comprehensive training for the users must be done to ensure that the users will know how to operate the system properly.

- Compliance with Regulations
    - The system must follow the rules and regulations of NU-APC standards.
- Data Privacy
    - Must comply with data privacy regulations of the Philippines or the Data privacy Act of 2012

5.4 Software Quality Attributes

The following are the software quality attributes that can be found in our system to ensure that the users will have a nonproblematic experience:

1. **Functionality**
    - The system should be accurate and will perform its intended functions reliably even if it is under various conditions

2. **Reliability**
    - System will be able to consistently perform its required functions without failure
    - In the event of a crash or failure, automatic recovery will be done within an hour.
    - System will have a high mean time between failures (MTBF) to ensure a continuous operation of the system.

3. **Usability**
    - The usability of the system will be user-friendly and intuitive to ensure that the users will not have problems interacting with it.
    - Training will be provided to make sure the users will be able to operate the system with minimal effort.

4. **Efficiency**
    - The system will be using resources effectively, including processing power and the storage.
    - It will be able to process data quickly and efficiently to provide real-time results.

5. **Maintainability**
    - The system should be easy to maintain and update.
    - This will allow the users to replace or even upgrade the system at will.

6. **Security**
    - The system will be able to protect against unauthorized access and data breaches.
    - This will include implementing strong encryptions, access controls, and regular security updates.

7. **Scalability**
    - The system will be able to handle increased loads to make sure the data does not overflow.

8. **Interoperability**
    - It will be able to work with other systems found in NU-APC such as security systems or even data storage.

Appendix A: Glossary

1. **DCV:** Dress Code Violation

2. **RBAC:** Role-Based Access Control

3. **Authentication:** The process of verifying a user's identity to access a system

4. **QR Code:** Quick-Response Code is a machine-readable code that encodes information about the students. Used for quick identification during violation issuance.

5. **Interoperability:** The ability of the system to integrate and exchange data with other systems or platforms.

6. **Framework:** Is a foundation used to build software applications, it usually provides pre-written code, tools, and libraries to simplify development process

7. **OS:** An Operating System is a program that manages a computer's hardware and software which enables applications to run.

8. **Stimulus:** This represents the triggering action or event that prompts the system to perform a specific function

9. **Response:** This is the system's reaction to the stimulus, or the action initiated by the user.